Experiment 7 - The Hall Effect

Daire O'Brien - 24365676

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1 Background

1.1 Discovery

In 1879, based on work by André-Marie Ampére and James Maxwell, Edwin Hall investigated how magnetic fields interacted with an electrical current in a conductor. He proposed that

$$E' \propto Mv$$

where $E' \equiv V$, $M \equiv |\vec{B}|$ and "v is the velocity of the electricity in the [object]" (E. H. Hall, 1879)[1] (At the time of Hall's paper the electron had only been theorized and wasn't discovered until 1897). Later, the equation was found to be

$$V_H = \frac{I_x B_z}{n_e C_e t}$$

where I_x is the current through the conductor, B_z is the magnetic field, n_e is the number of electrons, C_e is the charge of the electron and t is the thickness of the material. (Manny Soltero, 2023)[2]. The term $n_e C_e t$ represents the total charge density of the object.

The Hall effect describes how electrons accumulate on one side the material in the presence of a magnetic field.

1.2 Hall Effect Sensors

Hall effect sensors typically combine a Hall element and an op-amp. The Hall element has a similar structure to a Wheatstone bridge (R. Wei et al, 2017)[3]. The Hall voltage can be measured accross pins 1 and 3 when a bias voltage is applied between pins 2 and 4(Fig 1). If all of the resistors are not properly balanced the output voltage will be offset by some amount. In a larger IC the op-amp is used to amplify the voltage to levels that can be read by a computer.

Hall effect sensors are often used to measure the position or angle of something, for example a stick in a game controller, the rotation speed/postion of a motor's shaft or measuring current. Typically, magnets are mounted to one of the moving components and the sensor on the other side. If multiple magnets are used then the velocity of the object can be measured by maasuring the "bumpy" magnetic field. If the device itself can be calibrated or started from some known position then Hall effect sensors can be used to determine the position of the device to a comparitively high accuracy.

As Hall effect sensors are solid state components they can often be used in areas were mechanical wear of a more traditional component may be a problem, or where there might be a physical disconnect from the sensor and what it's trying to measure.

2 Method

1. The apparatus was wired together. The voltmeter was chained in series with the top plug of the Hall effect waver, with the bottom of the wafer going back into the negative end of the power supply. The electromagnet was connected to the second power supply.
2. The sensor PSU was turned on and set to 0 V. The electromagnet PSU was set to 1 A. The values on the ammeter and voltmeter were recorded.

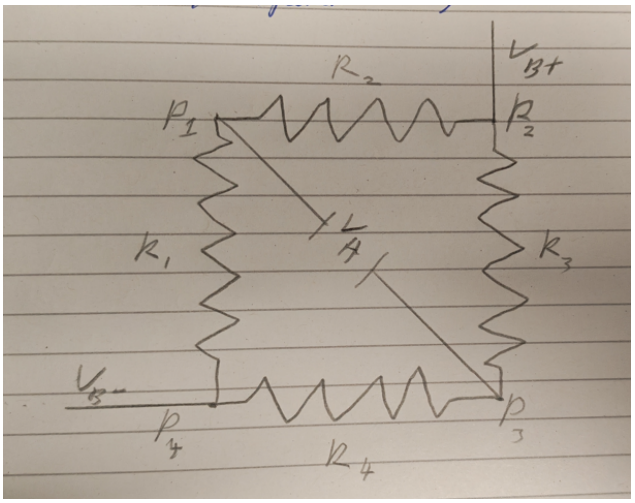


Figure 1: Diagram of a Hall Element

3 Results

$$T = 1\,\text{mm}$$
$$R_H = \frac{V_H T}{IB}$$
$$R_{H(1\,\text{A})} = \frac{-}{a}$$

4 References

1. Edwin Hall, 1879, *On a New Action of the Magnet on Electric Currents*, American Journal of Mathematics Vol. 2 No. 3

2. Manny Soltero, 2023, *What Is a Hall-effect Sensor?*, Texas Instruments technical article

3. Wei, Rongshan & Guo, Shizhong & Yang, Shanzhi, 2017, *Low-Power CMOS Integrated Hall Switch Sensor*, Active and Passive Electronic Components. 2017. 1-9. 10.1155/2017/5375619.